

Ayush Mittal

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EXPERIENCE

PESU C-ISFCR

Jun 2026 – Aug 2026

Research Intern — AFEM: Agentic Forensic Evidence Model

Bengaluru, India

- Developed a **five-phase deterministic digital forensics framework** for autonomous email agents covering evidence collection, integrity verification, timeline reconstruction, explainable attribution, and forensic reporting.
- Implemented **tamper-evident SHA-256 hash chaining** and explainable **AUTH/SCOPE/INJ/AMBIG attribution** for post-incident behavioural analysis.
- Validated on **1,000 Enron emails + synthetic evaluation data**, achieving **268/268 regression tests passed** and automated JSON/HTML investigation reports.

PROJECTS

Aero-FNO — Geometry-Aware Fourier Neural Operators | Python, PyTorch, FNO, SciPy

- Developed a **Fourier Neural Operator surrogate for aerodynamic CFD**, predicting pressure and velocity fields across **6,400+ simulations** with millisecond-scale inference.
- Introduced a **surface-weighted loss function**, reducing airfoil-surface pressure error from **143.5% to 34.7% (76% improvement)**.
- Demonstrated **~3,000× faster inverse aerodynamic optimization** than CFD, achieving **2–20% drag reduction** across test airfoils.

Dynamic Streaming via Kafka | Python, Apache Kafka, Flask, SQLite, REST APIs

- Engineered a **distributed real-time event-streaming platform** using Apache Kafka, Flask and SQLite, deploying broker, producer, consumer and admin services across **four nodes**.
- Implemented **dynamic topic provisioning, approval workflows, multi-user subscriptions and REST APIs** for real-time sensor, payment, weather and user-activity streams.
- Built an authenticated **Flask monitoring dashboard** with live topic/subscription statistics and operational logging for centralized management of distributed Kafka workloads.

Kickstarter Campaign Success Prediction | Python, LightGBM, MiniLM, Optuna, SMOTE

- Built a pre-launch ML pipeline over **~220,000 Kickstarter campaigns**, combining structured campaign features with **MiniLM text embeddings** to predict campaign success.
- Addressed class imbalance using **SMOTE** and optimized LightGBM with **Optuna**, achieving **70.8% accuracy and 0.774 ROC-AUC** on held-out data.
- Developed a soft-voting ensemble of **LightGBM, Random Forest and Logistic Regression**, achieving the highest evaluated **F1 score of 0.631**.

EDUCATION

Pes University

Aug. 2023 – May 2027

Bachelor of Technology in Computer Science

Bengaluru, India

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL, HTML/CSS

AI/ML: PyTorch, Scikit-learn, LightGBM, Pandas, NumPy, Ollama

Frameworks: LangGraph, React, Node.js, Flask, Express.js

Databases: MySQL, MongoDB, SQLite

Developer Tools: Git, Docker, Apache Kafka, REST APIs